SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Computer Science and Engineering

ASSESSMENT TOOL 1 – INTERACTIVE TREE SIMULATOR

Course Code:	CSA0305	**Course Title:**	Data Structures
CO Assessed:	CO3 – Precision (BL3, BL5)	**Assessment:**	Assessment Tool 1 – Interactive Tree Simulator
Weightage:	30% of CIA	**Total Marks:**	100
CO3 Statement:	*Solve problems for optimized data storage and retrieval using Binary Search Trees, AVL Trees, BTrees, Red-Black Trees, and Splay Trees.*		
Student Name:	M.Deepika	**Reg. No.:**	192372296
Section / Batch:	2023	**Date:**	29/07/2026

Code of Conduct

I _______M.DEEPIKA-192372296________ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action. Signature of the Student: _____deepika.m__

Instructions

- This assessment tool contains 5 Interactive Tree Simulator questions, Total: 100 Marks.
- Students can use any online/offline Interactive Tree Simulator. Demonstrate insertion, deletion, searching, and traversal operations wherever applicable.
- When a student answer using simulator, in evaluation the answer should have captured screenshots after every major operation.
- Submit the report in PDF format with properly labelled screenshots as proof of completion.

Section / Topic	Focus	Q Nos	Marks

Binary Search Tree	Properties, binary tree and binary search tree, insertion and deletion	1	20
Tree Traversals	Traversal types, In order, Post order and Pre order	2	20
AVL Tree	Balancing factor, Rotation, Types of rotation, examples	3	20
B Tree, 2-3 tree, 2-3-4 tree	Properties, structures and Operations	4	20
Red Black Tree	Properties, Example and Operations	5	20
		GRAND TOTAL:	**100 Marks**

Annexure A – Interactive Tree Simulator

1. Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80 Perform:
    a. Inorder Traversal
    b. Search for 60
    c. Delete 30
    d. Display the final BST.
2. Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80. Find all the traversals.
    a. Inorder
    b. Preorder
    c. Postorder
3. Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.
4. Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75
5. Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree. Analyze their heights.

Rubrics for Calculation

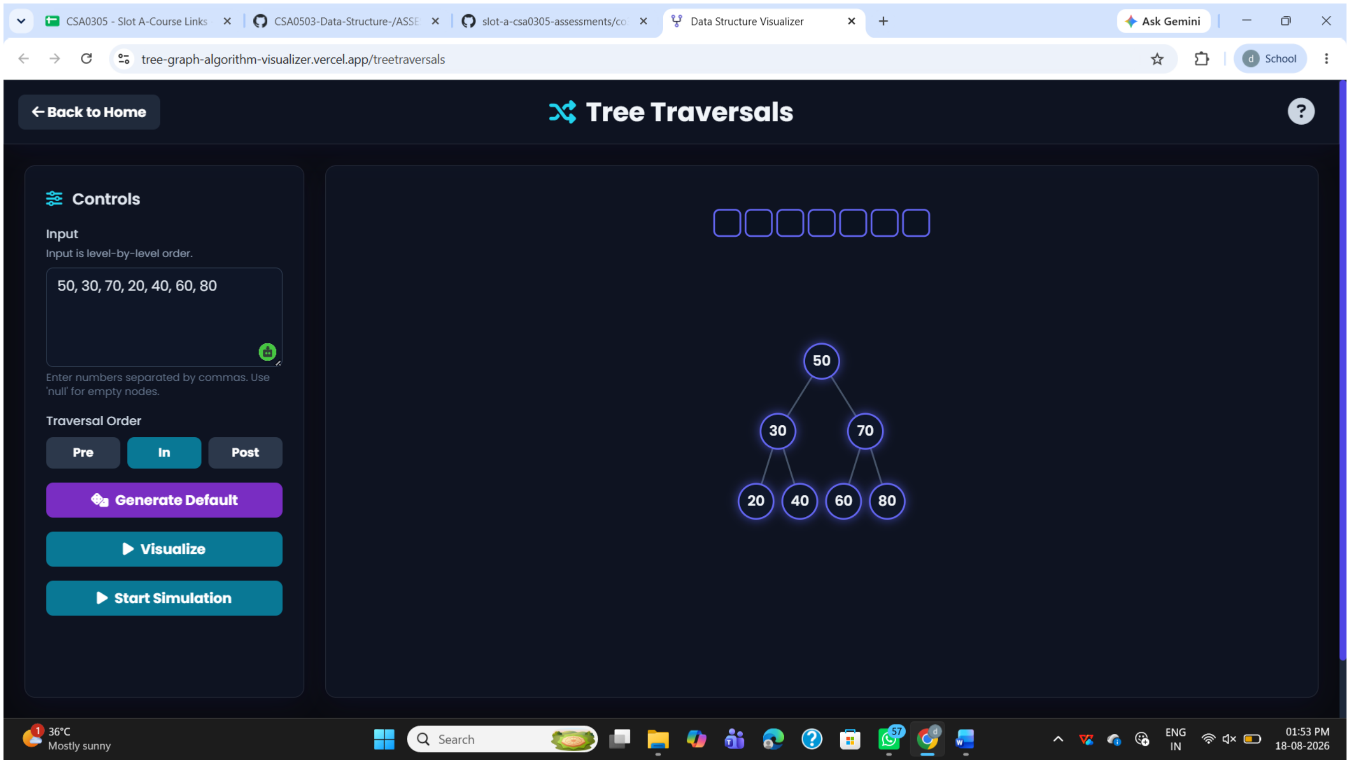
Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Correct Tree Construction	20	Tree is constructed correctly with all operations performed accurately	Minor errors in construction	Some operations incorrect	Tree construction mostly incorrect
Understanding of Tree Operations	20	Clearly explains insertion, deletion, search and balancing operations	Explains most operations correctly	Limited understanding	Unable to explain operations

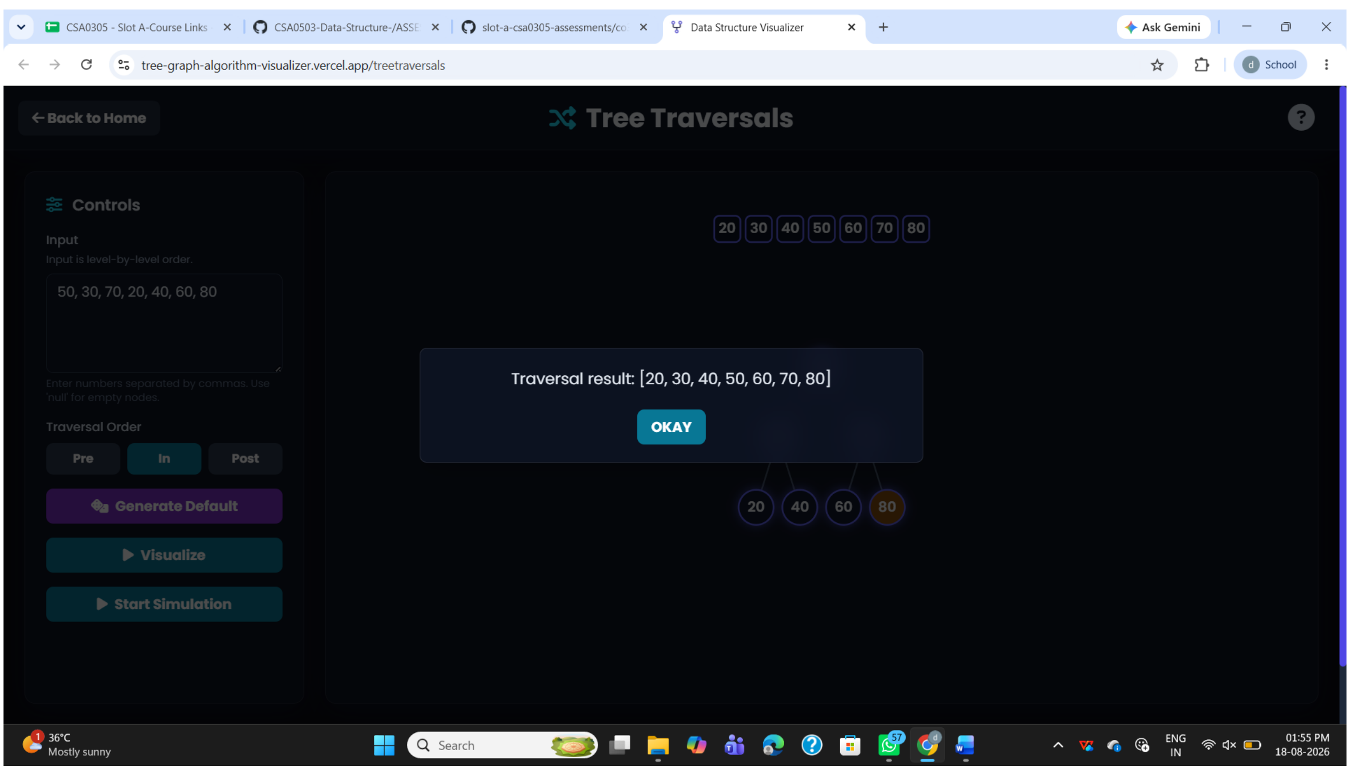
Analysis of Tree Behaviour	30	Correctly analyses balancing, rotations, node split/merge, splaying	Good analysis with minor omissions	Partial analysis	Poor or incorrect analysis
Presentation	20	Well-organized report with accurate observations and conclusion	Good report with minor issues	Basic report with limited observations	Incomplete report
Accuracy of Responses	10	90–100% correct answers	75–89% correct answers	50–74% correct answers	Below 50% correct answers

Faculty In-charge	**Course Coordinator**	**Program Director**
Signature: ______________	Signature: ______________	Signature: ______________
Date: _________________	Date: _________________	Date: _________________

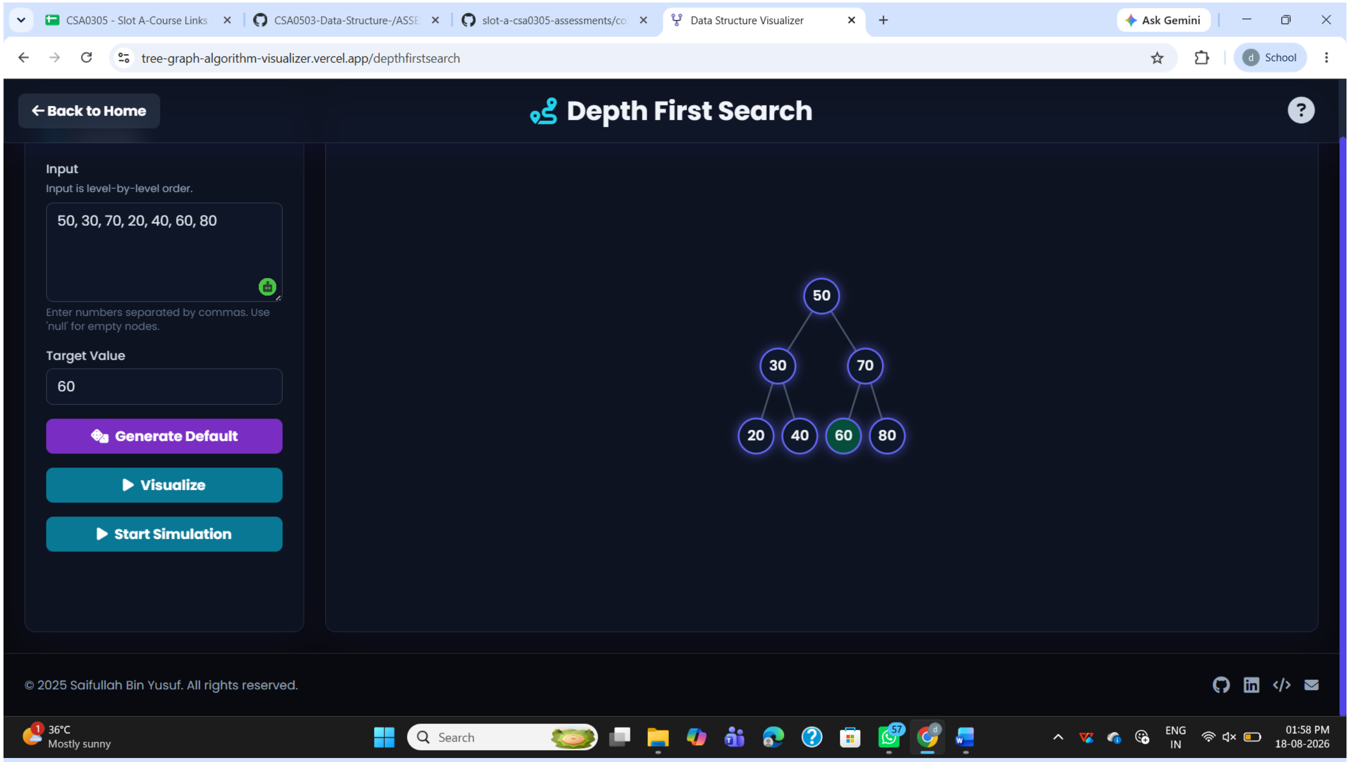
1. Construct a Binary Search Tree (BST) by inserting the following keys: 50, 30, 70, 20, 40, 60, 80 Perform:

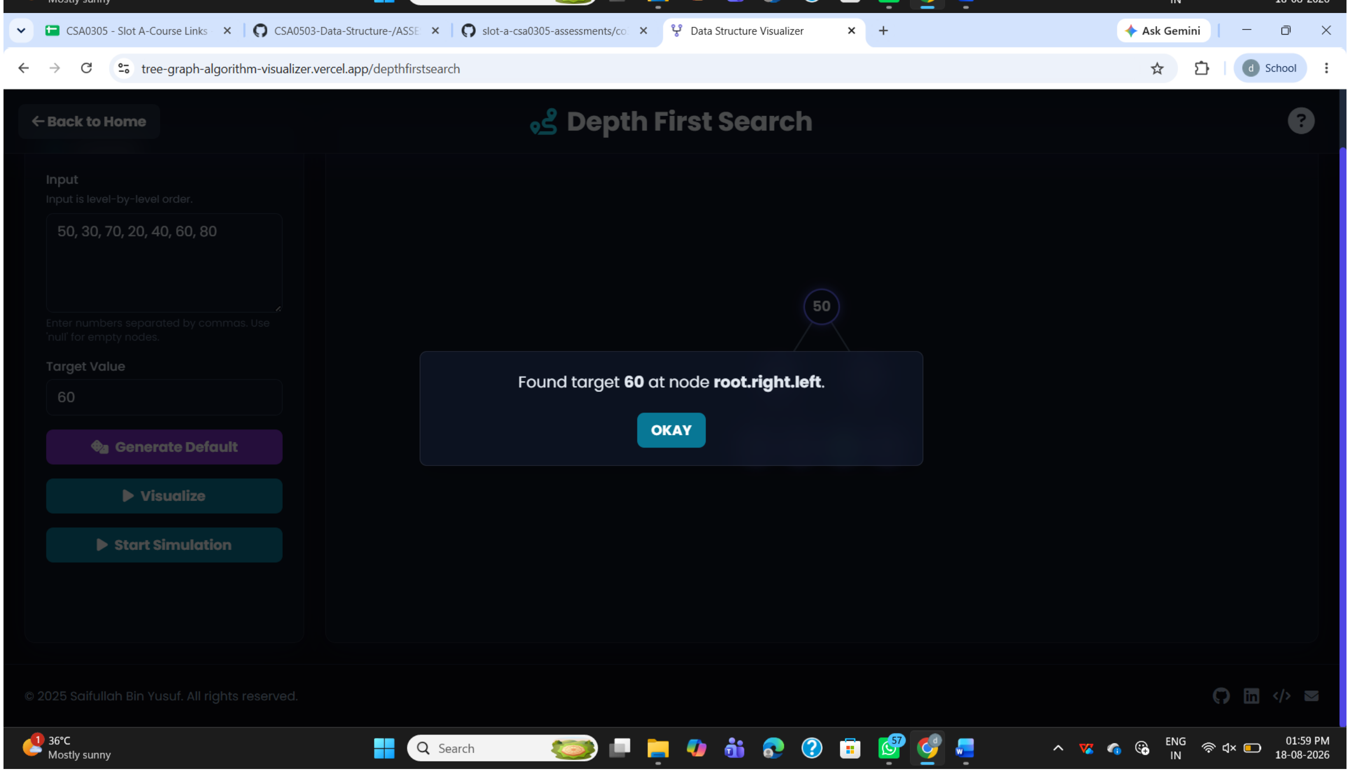
a. Inorder Traversal



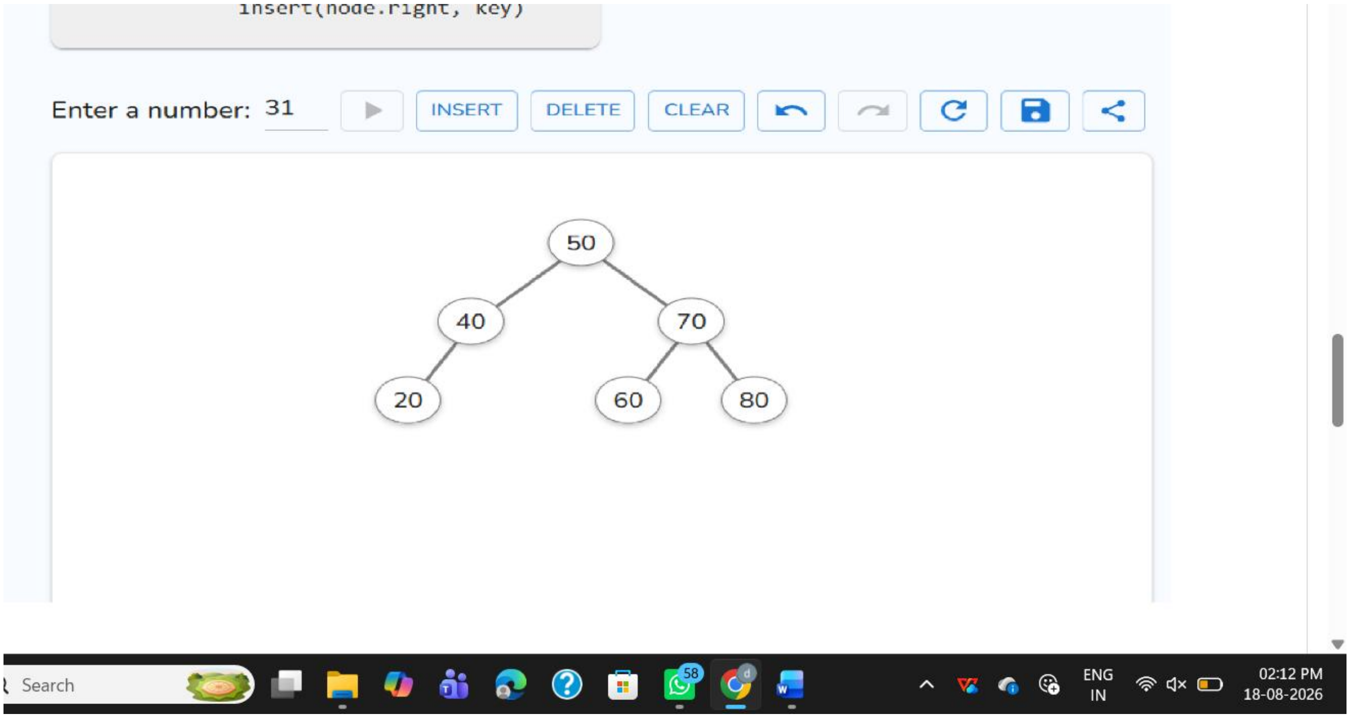


b.Search for 60

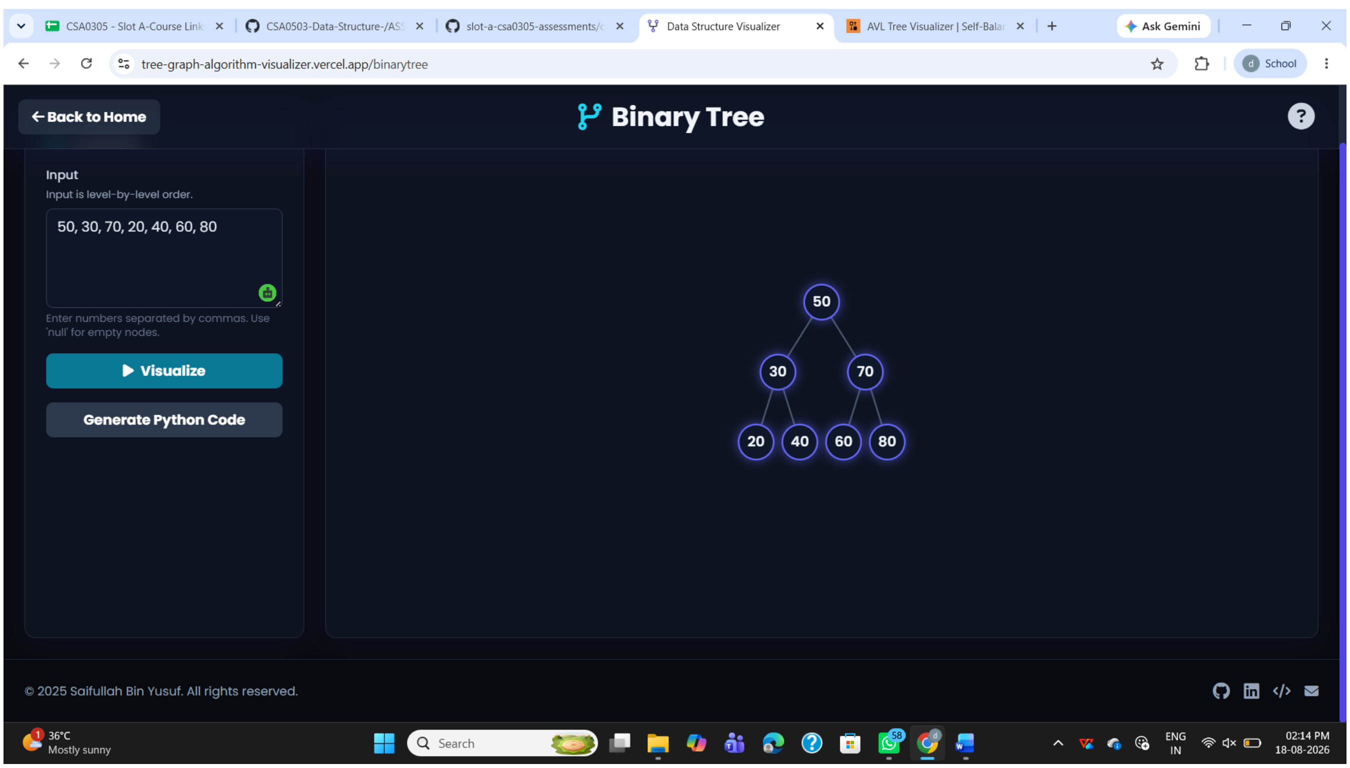




C. Delete 30

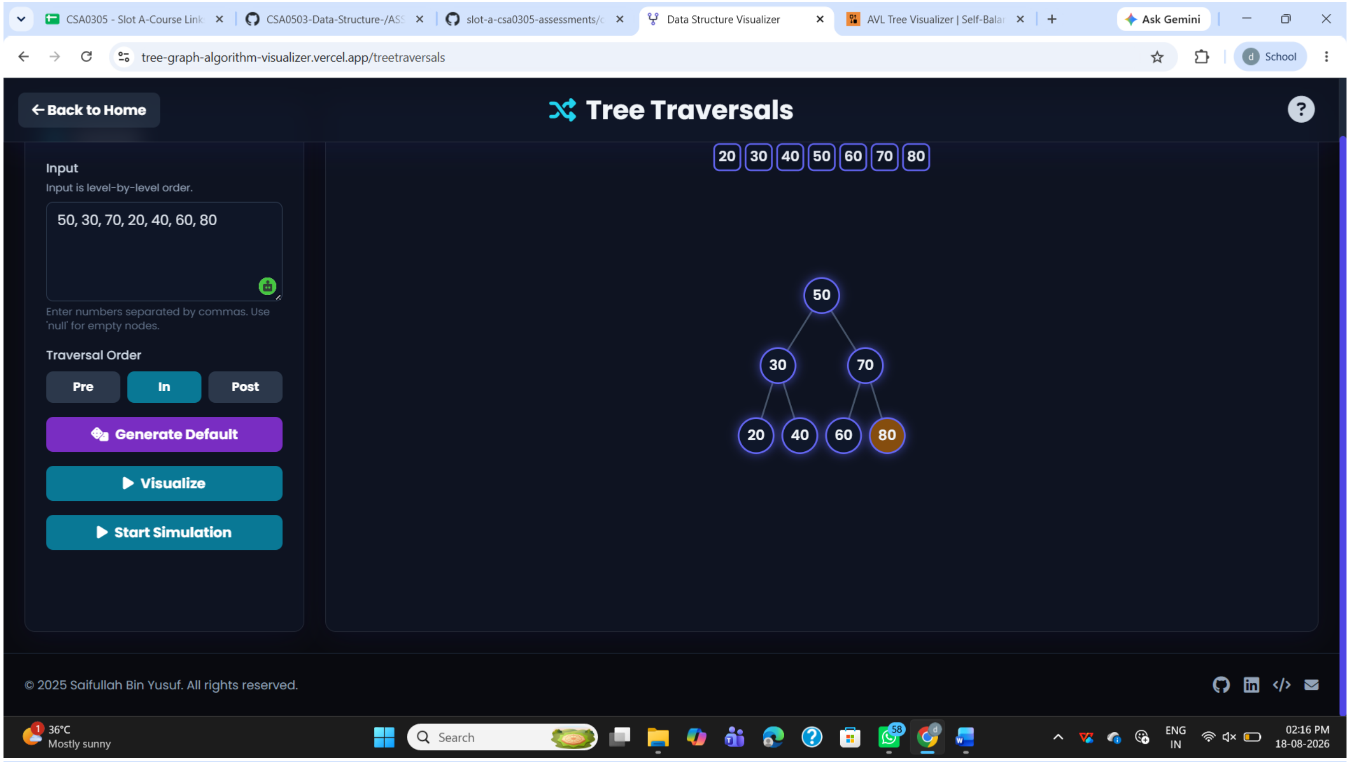


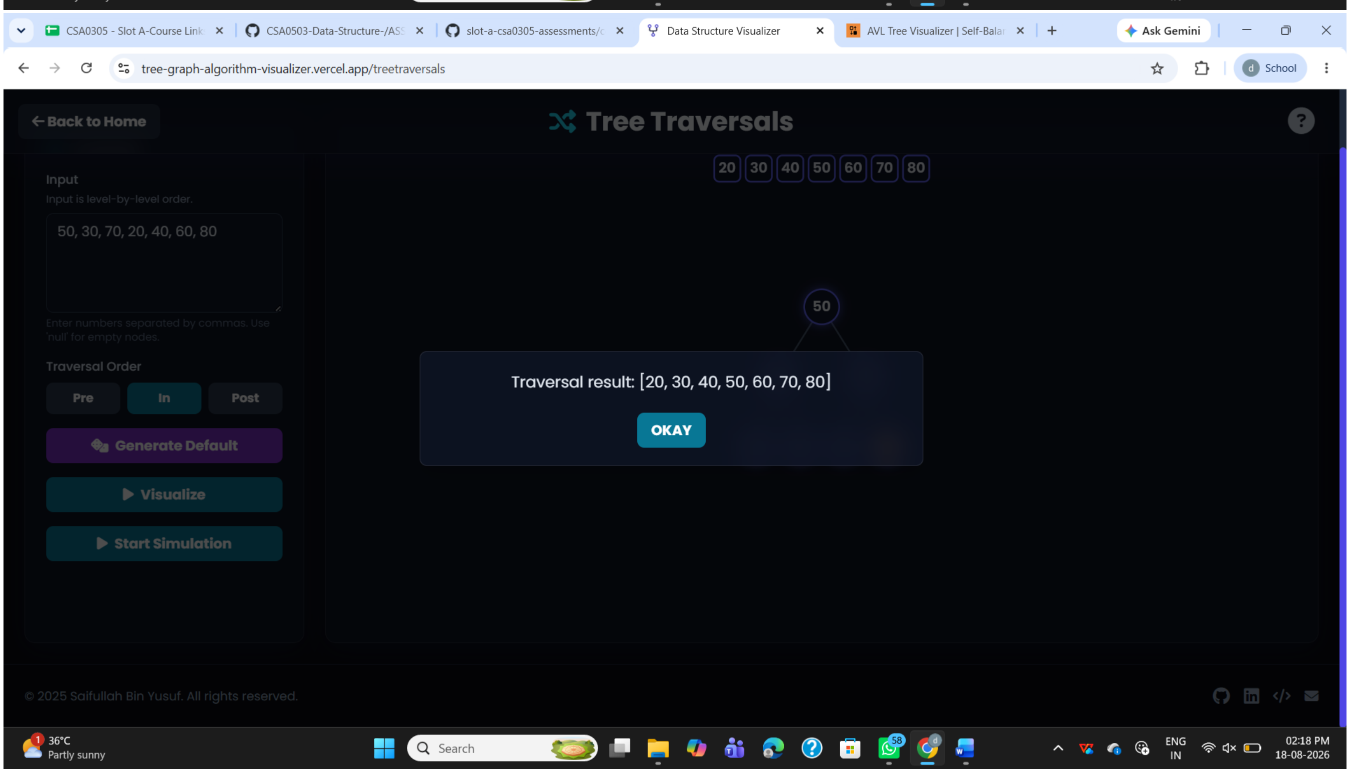
d.Display the final BST.



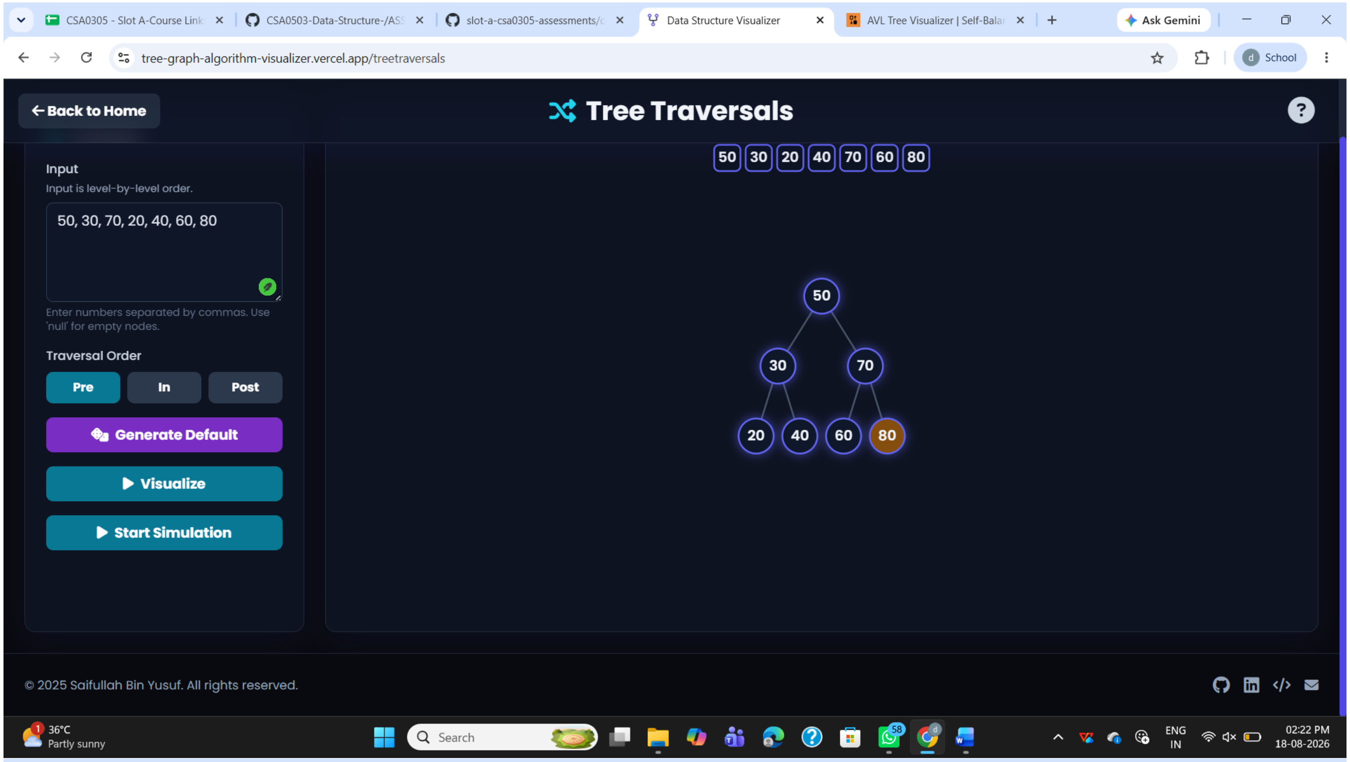
2.Complete Tree Traversals by Inserting 50, 30, 70, 20, 40, 60, 80 Find all the traversals. a. Inorder

a. Inorder





b. Preorder


tree-graph-algorithm-visualizer.vercel.app/treetraversals
Back to Home
Tree Traversals
50 30 20 40 70 60 80
Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
50
30
70
20
40
60
80
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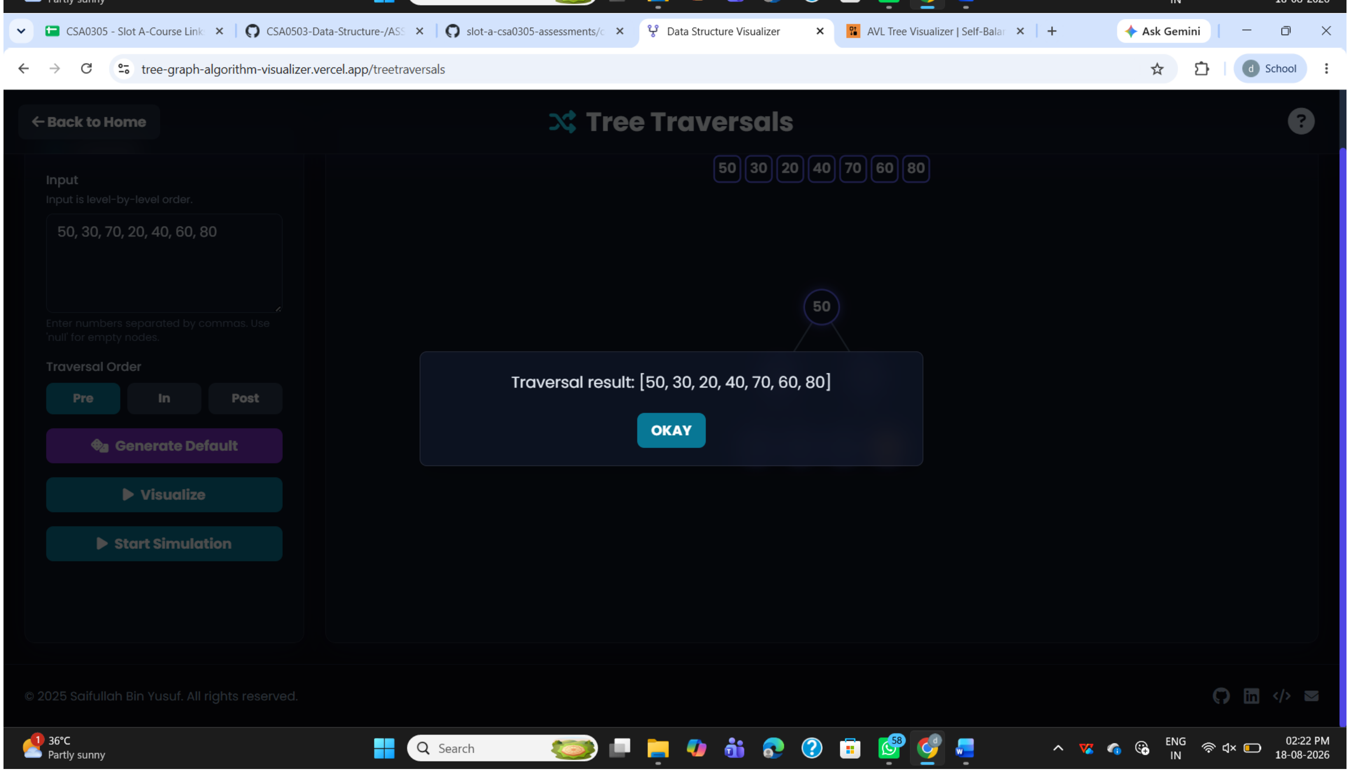
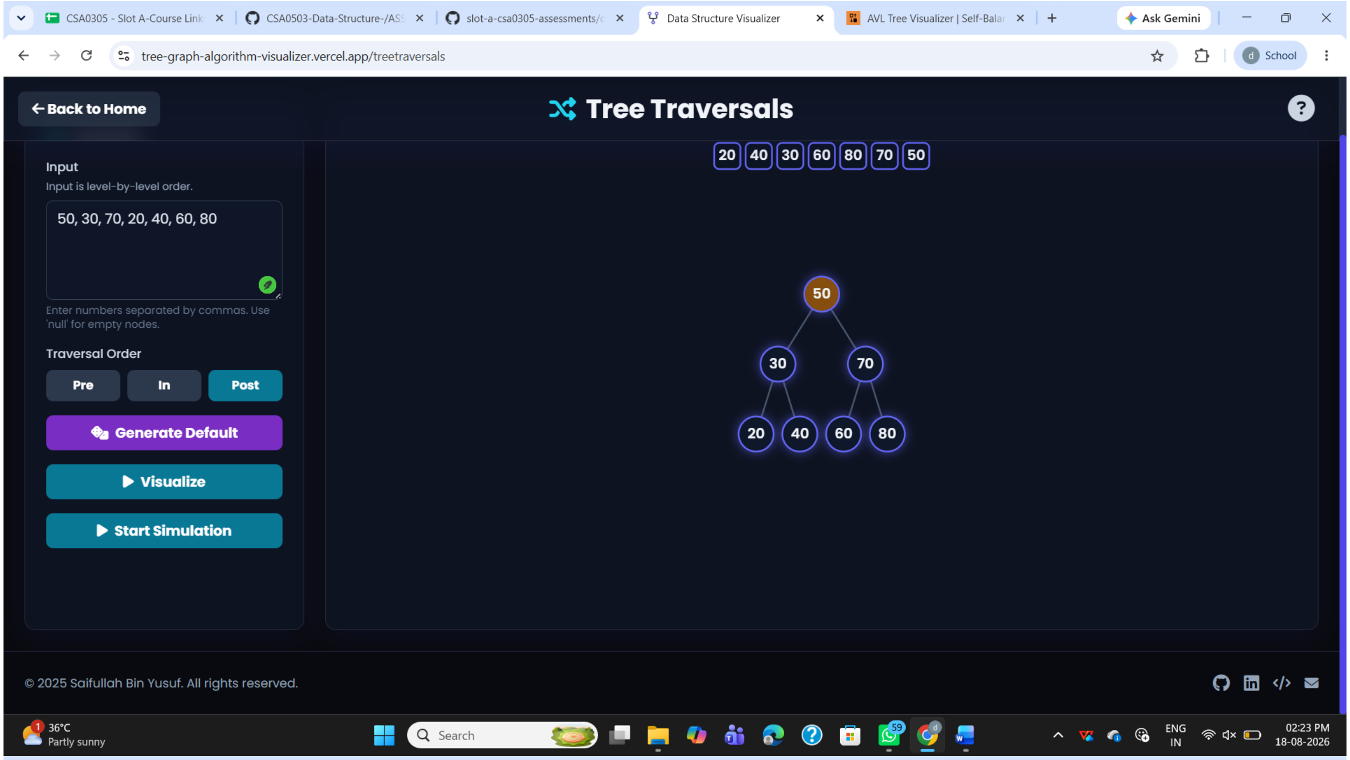

tree-graph-algorithm-visualizer.vercel.app/treetraversals
Traversal result: [50, 30, 20, 40, 70, 60, 80]
OKAY

c. Postorder


Tree Traversals
Back to Home
Input
Input is level-by-level order.
50, 30, 70, 20, 40, 60, 80
Enter numbers separated by commas. Use 'null' for empty nodes.
Traversal Order
Pre
In
Post
Generate Default
Visualize
Start Simulation
20 40 30 60 80 70 50
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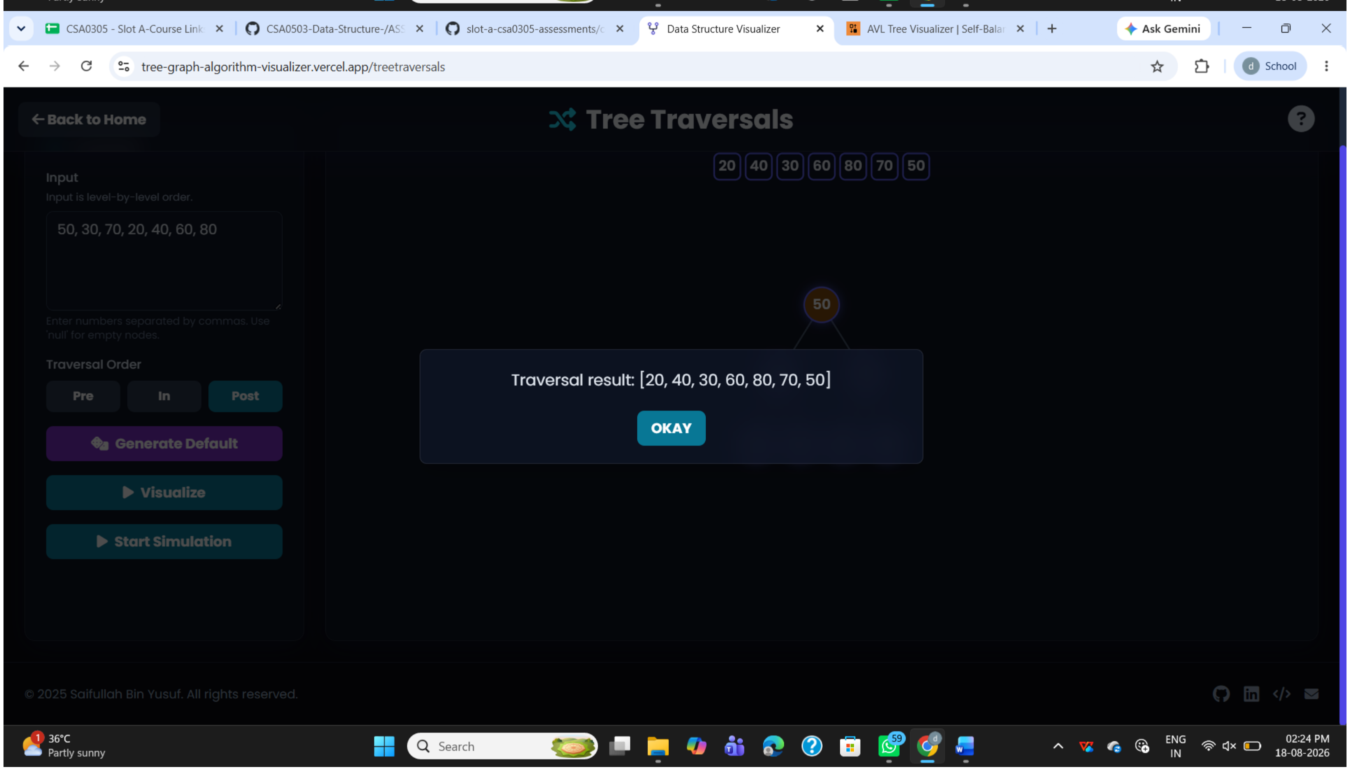
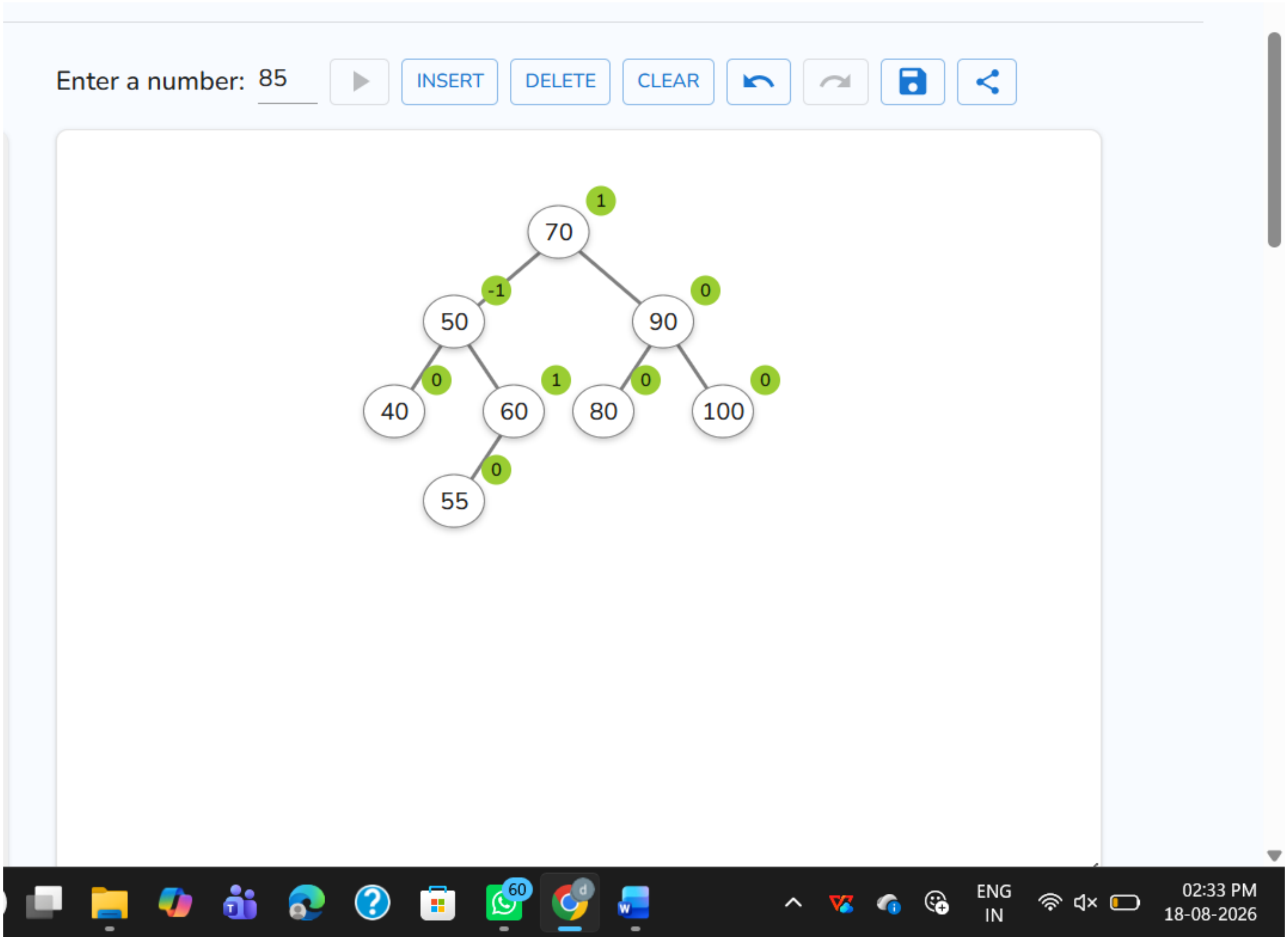
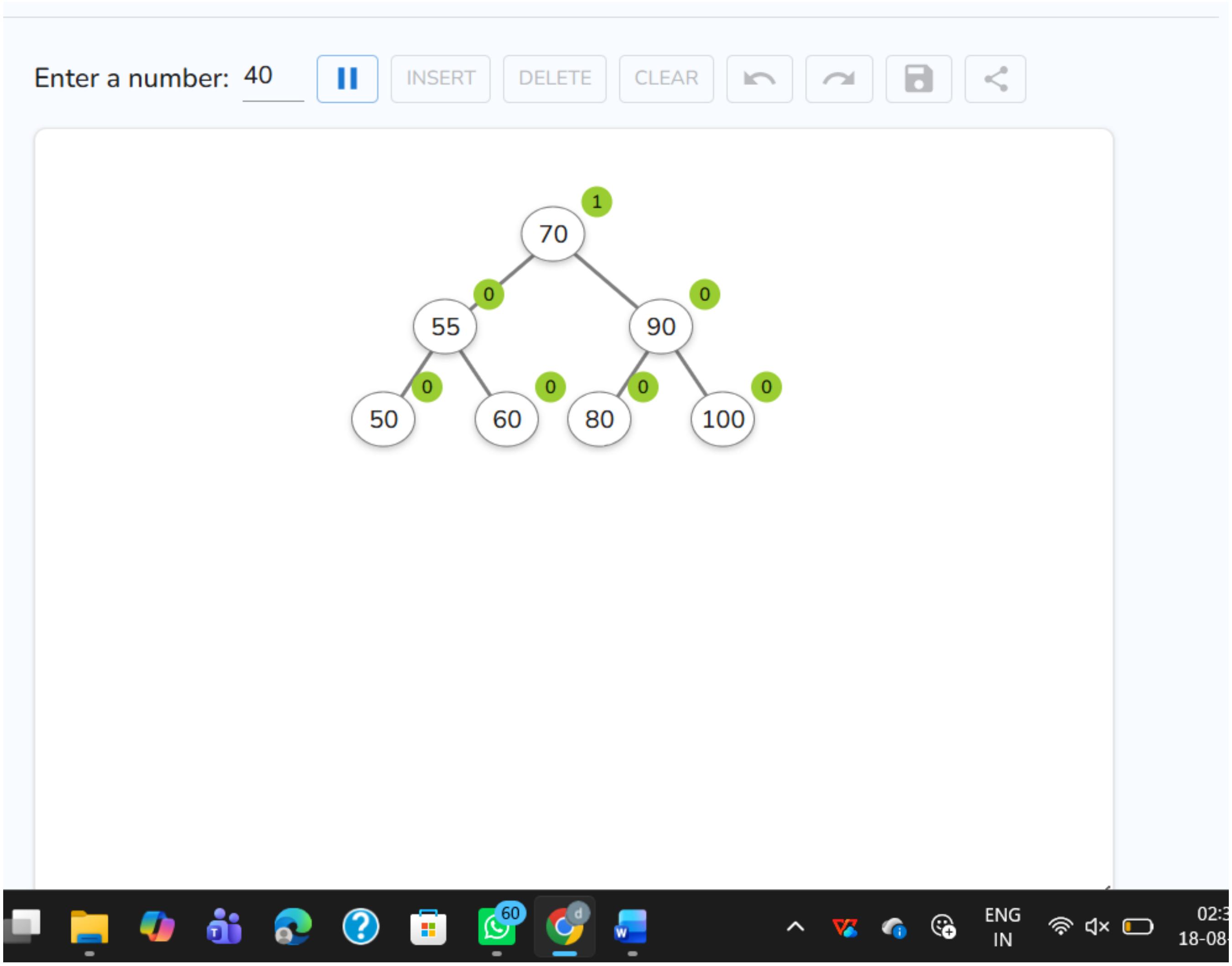
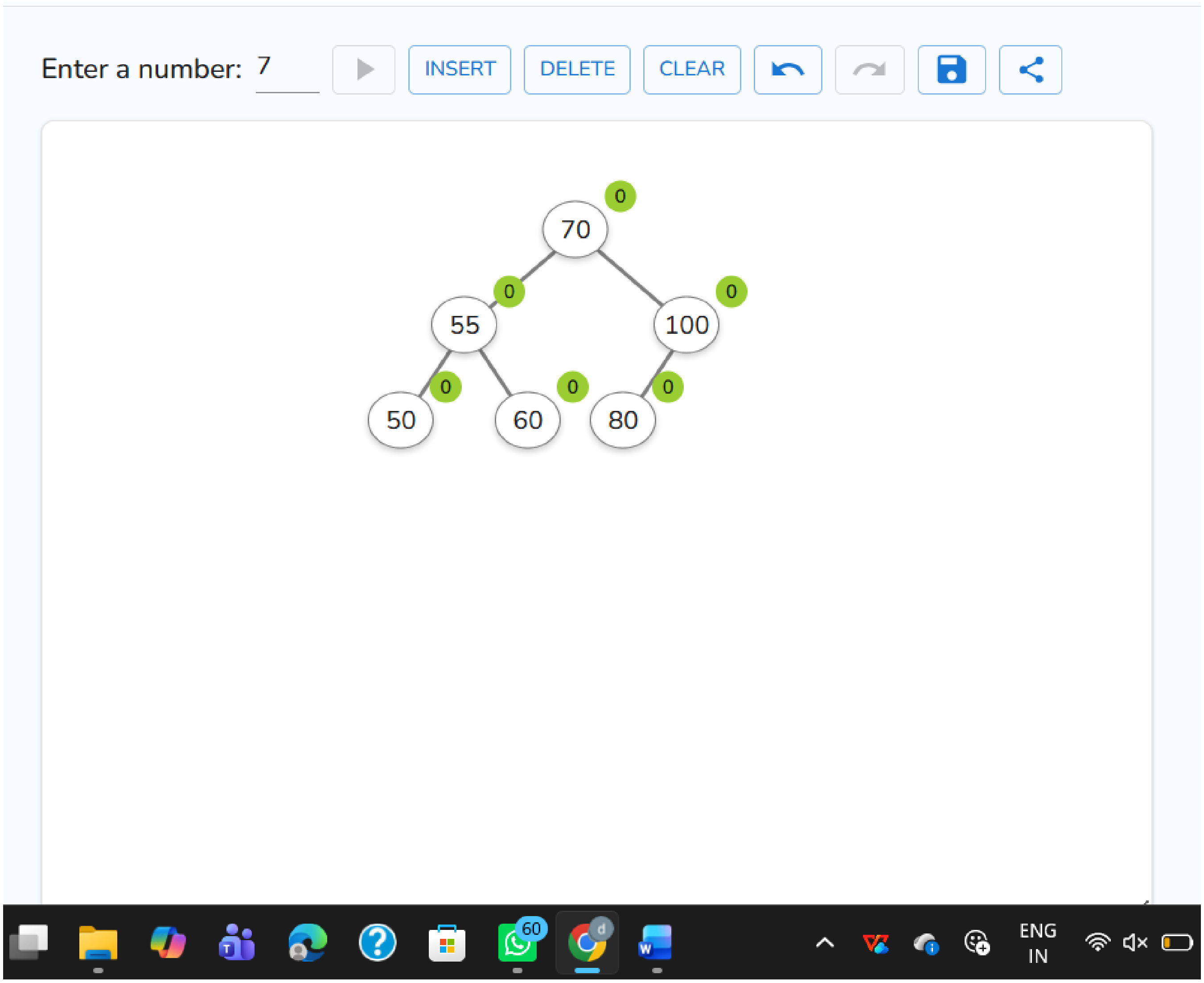

Traversal result: [20, 40, 30, 60, 80, 70, 50]
OKAY

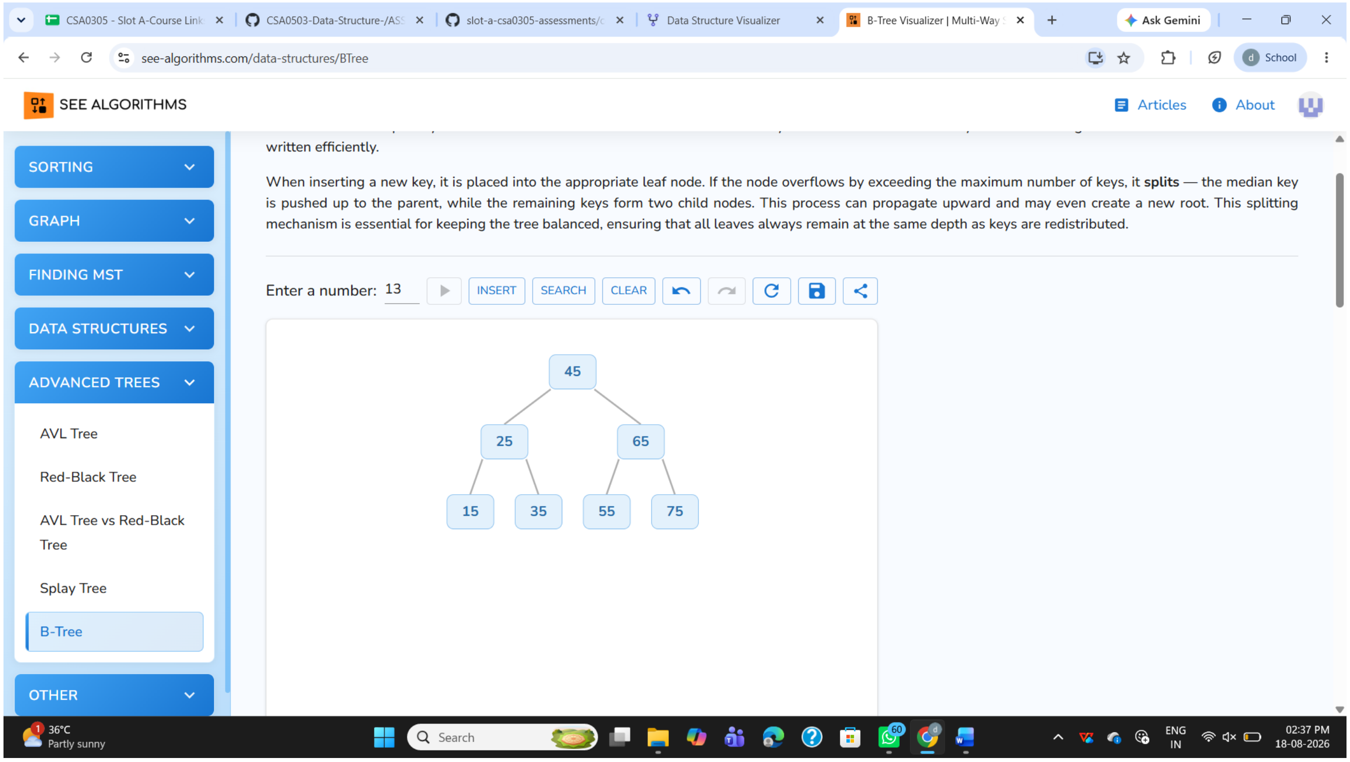
3.Insert 70, 50, 90, 40, 60, 80, 100, 55 into an AVL Tree. Delete 40 and 90. Analyze how the tree rebalances after each deletion.




Enter a number: 40
INSERT
DELETE
CLEAR
70
1
55
0
90
0
50
0
60
0
80
0
100
0
ENG
IN
02:3
18-08


Enter a number: 7
INSERT
DELETE
CLEAR
70
0
55
0
100
0
50
0
60
0
80
0
60
ENG
IN

4. Construct a B-Tree of order 3 by inserting 45, 25, 65, 15, 35, 55, 75



5.Insert 100, 80, 120, 60, 90, 110, 130 into a Red-Black Tree. Analyze their heights.